

Architecture Selection Justification - Smart Cities Municipal App

Architecture Selection: "We chose Layered Microservices Architecture for the Smart Cities Municipal Infrastructure Damage Reporting App."

Architectural Choice: Layered Microservices Architecture

Two Reasons:

- **Scalability:** A municipal app can have a high volume of traffic during severe weather events (e.g., storms causing multiple potholes or fallen trees). Microservices allow scaling specific components (like the Ticket Management Service) independently of the User Interface.
- **Fault Tolerance & Offline Support:** The architecture allows the Mobile App Component to queue reports locally while offline and sync them to the Ticket Management Service once connectivity is restored, without affecting the rest of the system.

Security Advantage: By adopting a layered microservices approach, sensitive citizen data (like GPS locations and contact info) is encapsulated within the Data Layer. It can only be accessed through authenticated API calls via the Business Logic Layer. This mitigates unauthorized data extraction and ensures strict access control between Citizens and Municipal Engineers.

Performance Benefit: The separation of heavy operations, such as GPS metadata extraction and image processing, into dedicated microservices prevents the main reporting workflow from slowing down. Asynchronous ticket dispatching ensures citizens receive instant confirmation while backend routing happens in the background.